Cambridge Assessment International Education

# Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*6162688048*

**BIOLOGY** **9700/42**

Paper 4 A Level Structured Questions **October/November 2021**

**2 hours**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **28** pages. Any blank pages are indicated.


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**[Turn over**

## Section A

Answer **all** questions.

**1 (a)** The Bowman's capsule of a nephron is involved in ultrafiltration.

Fig. 1.1 is a diagram of part of a Bowman's capsule and glomerulus.

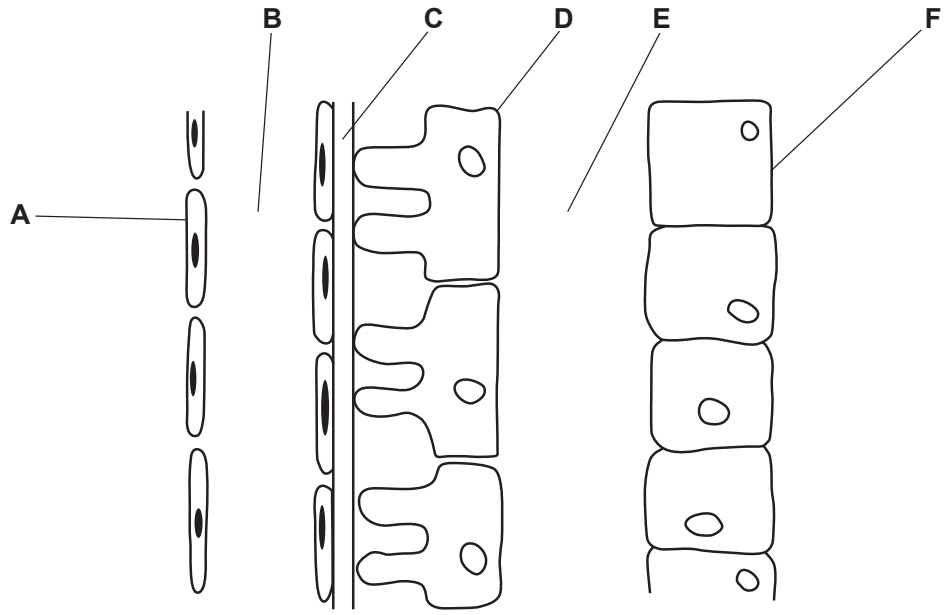



**Fig. 1.1**

With reference to Fig. 1.1, complete Table 1.1 using the letters **A** – **F**.

Each letter may be used once, more than once or not at all.

**Table 1.1**

| feature | letter |
|---|---|
| glomerular filtrate | ..... |
| basement membrane | ..... |
| podocyte cell | ..... |
| capillary endothelial cell | ..... |

[4]

**(b)** Describe **and** explain how the structures in the Bowman's capsule and its associated blood supply are adapted to allow ultrafiltration to take place.

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..... [5]

**(c)** The glomerular filtration rate (GFR) is the rate at which blood plasma is filtered in the Bowman's capsule.

Fig. 1.2 shows the relationship between GFR and mean renal arterial blood pressure.

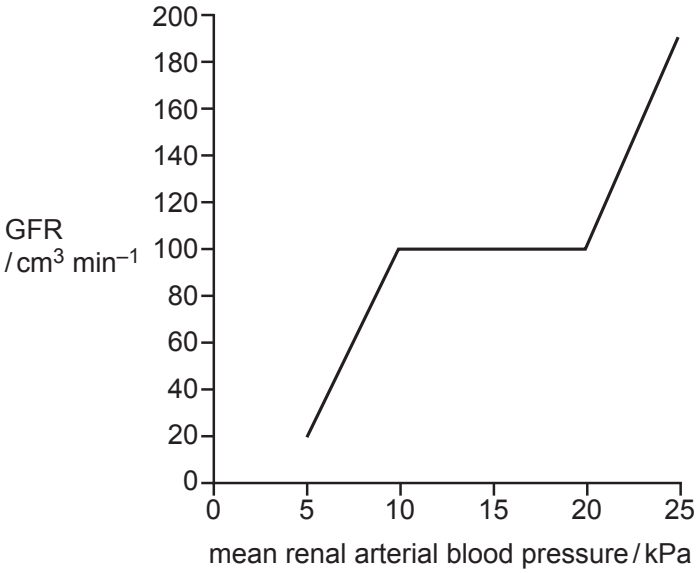



**Fig. 1.2**

**(i)** Comment on the relationship between GFR and mean renal arterial blood pressure.

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..... [2]

**(ii)** Suggest **one** reason why the GFR of a person might decrease.

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..... [1]

[Total: 12]

**Question 2 starts on page 6.**

**2** Scientists are researching new ways to reduce the global atmospheric carbon dioxide ($CO_2$) concentration. There are concerns that an increasing atmospheric $CO_2$ concentration may lead to effects that decrease biodiversity.

**(a)** Give **one** example of a human activity, other than deforestation, that contributes greatly to the increase in global atmospheric $CO_2$ concentration.

..... [1]

Algae are aquatic photosynthetic protoctists. Some researchers genetically modified the unicellular alga, *Chlorella vulgaris*, to try to increase the rate of the light independent stage of photosynthesis.

*C. vulgaris* was modified to increase the expression of the gene coding for aldolase. Aldolase is an enzyme that causes an increase in the concentration of rubisco.

Two cultures of *C. vulgaris*, one that was not genetically modified (unmodified) and one genetically modified, were grown under controlled conditions for 14 days. Samples were taken from the cultures at regular intervals during the 14 days to obtain measurements of dry mass.

The results are shown in Fig. 2.1.

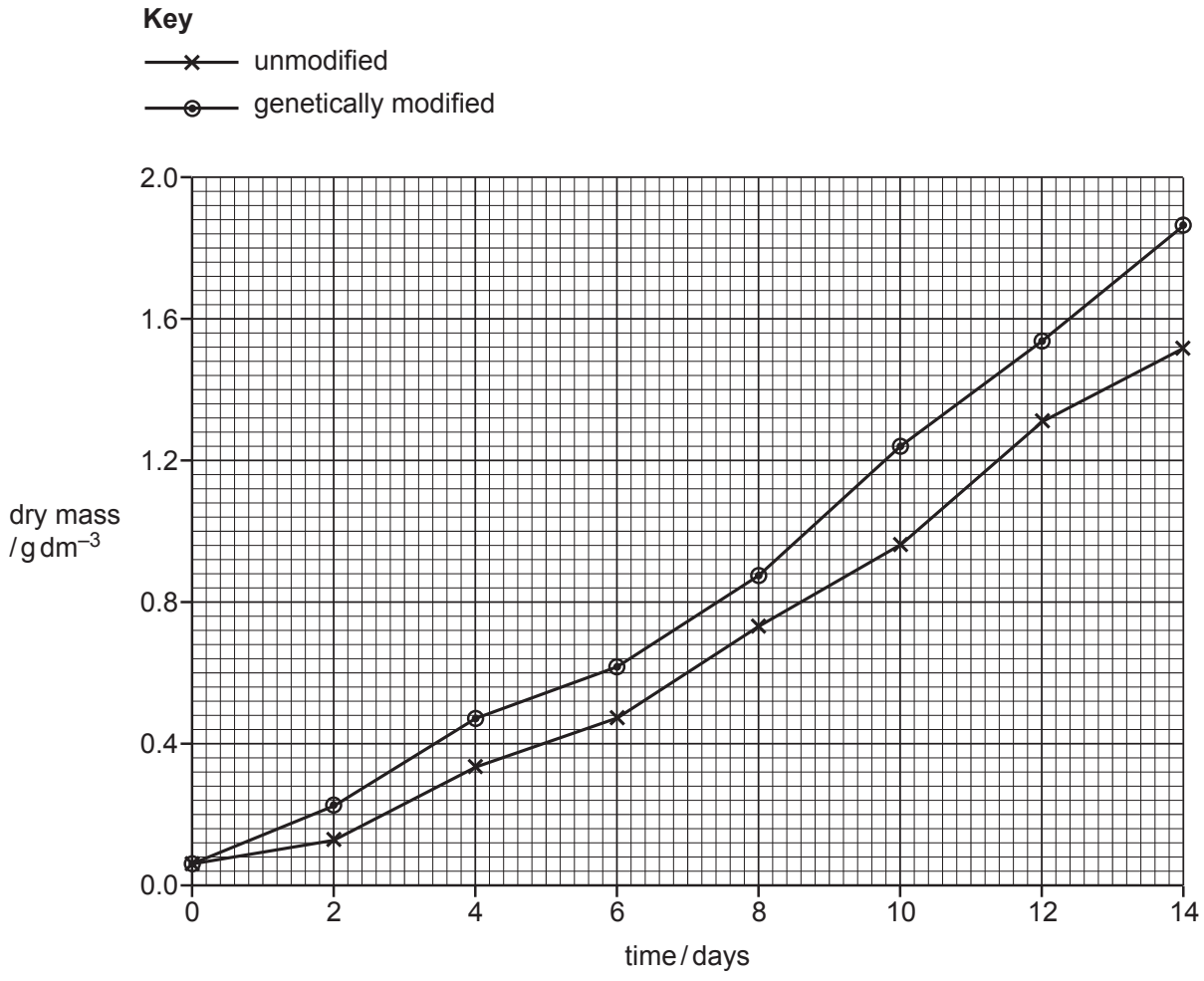



**Fig. 2.1**

**(b)** With reference to Fig. 2.1, describe the differences between the results for the two cultures.

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..... [3]

**(c)** Explain how the Calvin cycle was affected by the genetic modification of *C. vulgaris*.

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..... [3]

**(d)** Intermediate products of the Calvin cycle are needed to produce organic molecules for use by the cell.

Describe how these organic molecules are used by cells.

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..... [3]

**(e)** Planting large numbers of trees is one way to reduce global atmospheric $CO_2$ concentration. Large scale culture of genetically modified *C. vulgaris* could also reduce global atmospheric $CO_2$ concentration.

Suggest **one** advantage of using genetically modified *C. vulgaris* instead of trees to reduce global atmospheric $CO_2$ concentration.

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..... [1]

[Total: 11]

3 During the germination of barley seeds, amylase is produced.

**(a)** Describe the sequence of events that lead to the production of amylase during germination of barley seeds.

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..... [3]

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**(b)** Malting is a process involved in the production of a drink called beer. During malting, barley seed germination is controlled so that the sugars produced during germination can be used in the production of beer.

Fig. 3.1 shows two features of a germinating barley seed during the first five days of malting:
- activity of the amylase enzyme
- the percentage of starch reserves remaining in the barley seed.

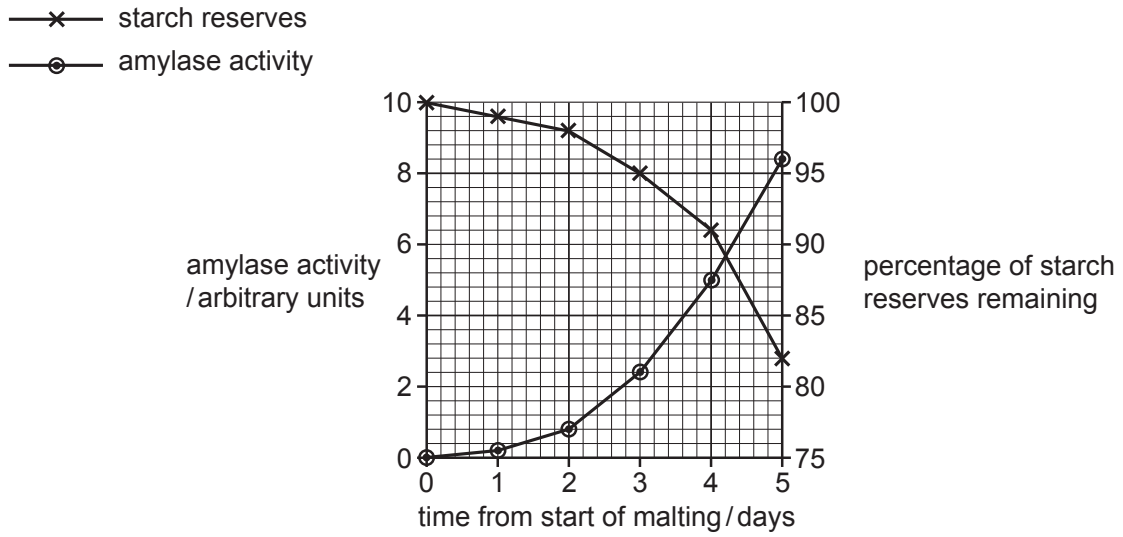



**Fig. 3.1**

**(i)** State the precise location of the starch reserves in the barley seed.

..... [1]

**(ii)** With reference to Fig. 3.1, describe **and** explain the effect of malting on amylase activity and the percentage of starch reserves remaining in the germinated barley seed.

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..... [4]

**(c)** In the malting process, germination is stopped before the concentration of sugars in the germinating barley seeds exceeds a concentration that causes shoot or root growth.

Drying the germinating barley seeds at 50 °C is one method used to stop malting.

**(i)** Explain how this method would stop malting.

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..... [3]

**(ii)** Suggest why malting is stopped before shoot or root growth occurs.

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..... [1]

[Total: 12]

**4** **(a)** Genome-wide association studies find links between single nucleotide polymorphisms (SNPs) and phenotypic features such as human diseases. SNPs are points on the DNA that vary in the population because of DNA base substitutions.

A genome-wide association study investigates the effect of genetic variation on a disease. A large number of people with the disease and a large number of healthy control individuals provide DNA. Microarray chips are used to identify the genotype of each individual at many SNPs.

The Wellcome Trust Case Control Consortium (WTCCC) study was an important genome-wide association study.

- The study used a microarray chip that identified each person's genotype at 500 000 different SNPs.
- The study looked for links between SNPs and 7 different diseases.
- For each disease, 2000 people with the disease were tested.
- Their results were compared with the results of 3000 healthy control individuals.

**(i)** Outline how microarrays are used in the analysis of genomes.

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..... [4]

**(ii)** Explain why bioinformatics was important to the WTCCC study.

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..... [2]

**(b)** Fig. 4.1 summarises results for three diseases in the WTCCC study. The 22 human autosomes and the X chromosome (chromosome 23) are shown.

Chromosome locations with SNPs that are associated with a disease at a statistically significant level (greater than 5 arbitrary units) are shown in black.

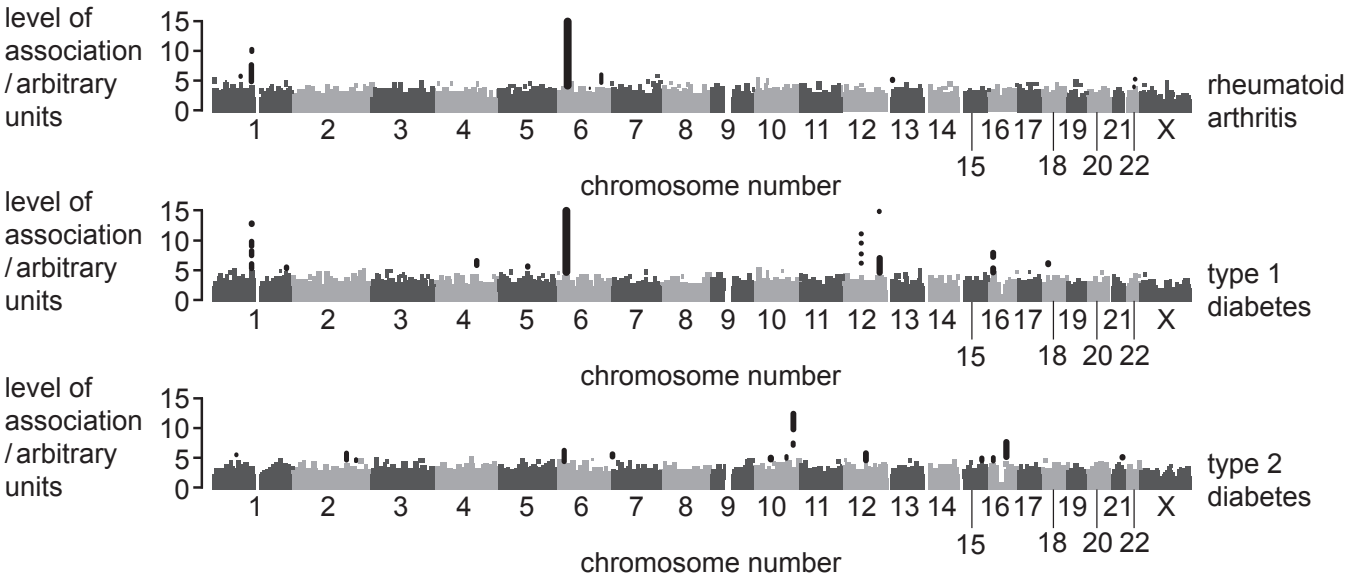



**Fig. 4.1**

**(i)** Identify the chromosomes that contain SNPs that have a high level of association with **both** rheumatoid arthritis and Type 1 diabetes.

..... [1]

**(ii)** With reference to Fig. 4.1, compare the genetic basis of the three diseases.

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..... [2]

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**(c)** Individuals can choose to have their DNA analysed on a microarray chip to predict their risk of developing different diseases.

Outline the social **and** ethical considerations of this type of DNA analysis.

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..... [3]

[Total: 12]

**5** Biodiversity can be considered at three different levels.

Fig. 5.1, Fig. 5.2 and Fig. 5.3 are images that show these three different levels of biodiversity.



**Fig. 5.1**



**Fig. 5.2**



**Fig. 5.3**

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**(a)** Describe the level of biodiversity most clearly visible in each image.

Fig. 5.1 .....

Fig. 5.2 .....

Fig. 5.3 .....

[3]

**(b)** Prairie strips are restored habitat areas planted with native grasses and wildflowers. Prairie strips are usually located between or around the edges of fields of maize or soy bean.

Students in North America wanted to investigate biodiversity along a prairie strip.

**(i)** Outline how the students could measure the biodiversity of plants and insects along a prairie strip.

plants .....

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.....

.....

insects .....

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.....

.....

[4]

**(ii)** Mathematical methods and statistical tests help to summarise and compare biodiversity data.

Name the mathematical method **or** statistical test that the students can use:

to estimate the total biodiversity of a prairie strip

.....

to test whether plant biodiversity is associated with insect biodiversity.

.....

[2]

**(c)** Prairie strips have environmental and economic effects.

Scientists have made long-term measurements of the environmental effects of prairie strips.

The scientists found that when prairie strips formed 10% of a crop-field:
- soil erosion decreased by 95%
- mineral loss from the field decreased by 90% for phosphorus compounds
- mineral loss from the field decreased by 85% for nitrogen compounds.

Crop plants need phosphorus (P) and nitrogen (N) to grow.

Evaluate the **economic** effects of prairie strips on farming.

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..... [3]

[Total: 12]

**Question 6 starts on page 18.**

6 **(a)** ATP is needed for many metabolic processes in living organisms.

**(i)** Describe the properties of ATP that make it suitable for its role as the universal energy currency.

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..... [4]

**(ii)** Suggest why ATP is needed for protein synthesis.

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..... [2]

**(b)** Fig. 6.1 is a diagram of mitochondrion.

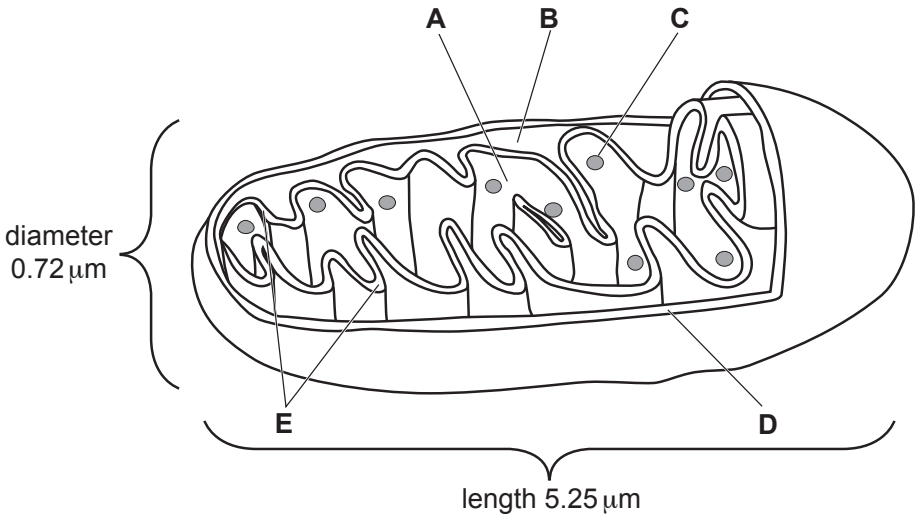



**Fig. 6.1**

**(i)** Complete Table 6.1 using the letters **A** to **E** from Fig. 6.1.

Each letter may be used once, more than once, or not at all.

**Table 6.1**

| statement | letter |
|---|---|
| the site of the Krebs Cycle | ..... |
| a phospholipid bilayer impermeable to $H^+$ ions | ..... |
| the site of translation | ..... |

[2]

**(ii)** Assume that the mitochondrion in Fig. 6.1 is a cylinder.

Calculate the surface area of this mitochondrion.

Use the formula: Surface area of cylinder = $2\pi r^2 + 2\pi rh$

Show your working.

surface area = ..... $\mu m^2$ [2]

**(iii)** The inner membrane of the mitochondrion has a much larger surface area than the outer membrane because of the presence of cristae.

Different cell types vary in the number of cristae present per mitochondrion. Cardiac muscle cells have mitochondria with a very large number of cristae.

Suggest **and** explain why cardiac muscle cells have mitochondria with very large numbers of cristae.

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..... [2]

[Total: 12]

**7** **(a)** The Labrador is a variety of domestic dog. Labradors have fur that can be brown, black or yellow.

In Labradors, *TYRP1* is one gene that codes for fur colour. This gene has two alleles, **B** and **b**.

- The dominant allele, **B**, codes for the enzyme tyrosinase that functions in the pathway to produce melanin, leading to black fur.
- The production of melanin in Labradors is very similar to the production of melanin in humans.
- The recessive allele, **b**, codes for an enzyme that results in the production of a brown form of melanin, leading to brown fur.

Outline how melanin may be produced in Labradors to produce black fur.

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..... [2]

**(b)** Another gene, *MC1R*, interacts with *TYRP1*.

*MC1R* has two alleles, **E** and **e**.

- The dominant allele, **E**, allows the alleles of *TYRP1* to be expressed.
- The recessive allele, **e**, prevents the alleles of *TYRP1* from being expressed.
- When no form of melanin is produced the Labrador will have yellow fur.

**(i)** Construct a genetic diagram to show the ratio of possible offspring from a cross between a black male Labrador, heterozygous for both genes, and a yellow female Labrador, heterozygous for *TYRP1*.

| | | | |
|---|---|---|---|
| parental phenotype | black | × | yellow |
| parental genotype | | | |
| gametes | | | |
| offspring genotypes | | | |
| offspring phenotypes | | | |

ratio .....

[6]

**(ii)** State the term used to describe a protein that is involved in the control of gene expression in eukaryotes.

..... [1]

[Total: 9]

**8** Twenty million years ago, an ocean covered the area where the country of Panama is now located. There was a gap between the continents of North America and South America through which the waters of the Atlantic and Pacific Oceans flowed freely.

The porkfish, *Anisotremus sp*, lived in this area between North America and South America.

Fig. 8.1 shows a porkfish.



**Fig. 8.1**

About 3 million years ago, volcanic activity and sedimentation formed a narrow strip of land, Panama, joining North America and South America.

Fig. 8.2 shows the area 20 million years ago and now.

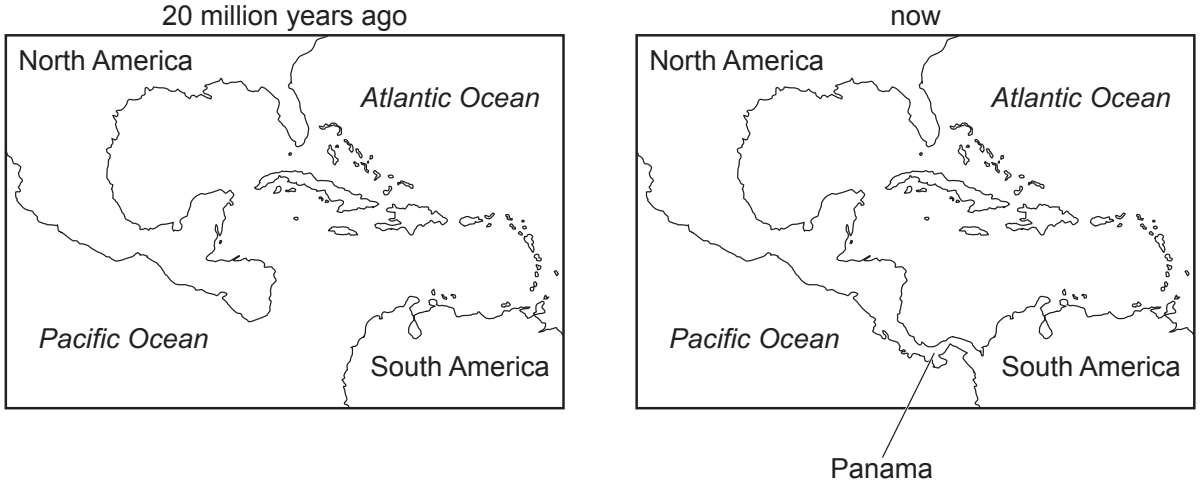



**Fig. 8.2**

Twenty million years ago, porkfish in the Atlantic and Pacific Oceans were able to breed successfully and produce fertile offspring.

Explain why Atlantic porkfish and Pacific porkfish are now **not** able to breed successfully to produce fertile offspring.

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..... [5]

[Total: 5]

## Section B

Answer **one** question.

**9** **(a)** Describe the roles of ADH **and** the collecting ducts in osmoregulation. [9]

**(b)** Describe the structure of a motor neurone. [6]

[Total: 15]

**10** **(a)** Explain how dip sticks function to test for glucose in a sample of urine. [7]

**(b)** Explain the control of gibberellin synthesis **and** outline how gibberellin stimulates stem elongation. [8]

[Total: 15]

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